QUIRINO HIGH SCHOOL

TECHPRO • GRADE 11 ACTIVITY SHEET

CSS IMAGE GALLERY & IMAGE SPRITES

Practical Activity • HTML and CSS

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Section: 11 - Bell _____ Score: _____

Learning Objectives

- Build an organized image gallery using HTML containers and CSS.
- Style gallery cards with borders, image sizing, captions, and hover effects.
- Explain how responsive layouts change at different screen widths.
- Use a CSS image sprite with background-image and background-position.
- Test, debug, and document a small web page in Visual Studio Code.

Before You Begin

Work in Visual Studio Code. You will create a responsive gallery and a navigation example using a sprite image. Use teacher-provided images or images you have permission to use. Keep image paths organized and descriptive.

Starter File 1 — index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>TechPro Image Lab</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header><h1>TechPro Image Lab</h1><p>Gallery and sprite practice</p></header>
  <main>
    <h2>Places to Explore</h2>
    <section class="gallery">
      <article class="gallery-card">
        <a href="images/place1.jpg" target="_blank"><img src="images/place1.jpg" alt="A scenic coastal
landscape"></a>
        <p class="caption">Coastal View</p>
      </article>
      <article class="gallery-card">
        <a href="images/place2.jpg" target="_blank"><img src="images/place2.jpg" alt="A green
forest"></a>
        <p class="caption">Green Forest</p>
      </article>
      <article class="gallery-card">
        <a href="images/place3.jpg" target="_blank"><img src="images/place3.jpg" alt="A mountain
landscape"></a>
        <p class="caption">Mountain Trail</p>
      </article>
      <article class="gallery-card">
```

```
        <a href="images/place4.jpg" target="_blank"><img src="images/place4.jpg" alt="A colorful night
sky"></a>
        <p class="caption">Night Sky</p>
      </article>
    </section>
    <h2>Navigation Sprite</h2>
    <nav class="sprite-nav" aria-label="Demo sprite navigation">
      <a href="#" class="sprite-icon home" aria-label="Home"></a>
      <a href="#" class="sprite-icon previous" aria-label="Previous"></a>
      <a href="#" class="sprite-icon next" aria-label="Next"></a>
    </nav>
  </main>
</body>
</html>
```

Starter File 2 — style.css

```
* { box-sizing: border-box; }
body {
  margin: 0; padding: 20px; font-family: Arial, sans-serif;
  color: #222; background: #f7f7f7;
}
header { padding: 18px; text-align: center; color: white; background: #6b1024; }
.gallery { display: flex; flex-wrap: wrap; gap: 12px; }
.gallery-card { width: calc(25% - 9px); border: 1px solid #ccc; background: white; }
.gallery-card:hover { border-color: #6b1024; }
.gallery-card img { display: block; width: 100%; height: auto; }
.caption { padding: 12px; margin: 0; text-align: center; }

/* Replace with the sprite image supplied/approved by your teacher. */
.sprite-icon {
  display: inline-block; height: 44px;
  background-image: url("img_navsprites.gif"); background-repeat: no-repeat;
}
.home { width: 46px; background-position: 0 0; }
.previous { width: 43px; background-position: -47px 0; }
.next { width: 43px; background-position: -91px 0; }

@media (max-width: 768px) {
  .gallery-card { width: calc(50% - 6px); }
}
@media (max-width: 480px) {
  .gallery-card { width: 100%; }
}
```

Guided Activity Steps

PART A — Set up the project

1. Create a folder named CSS_Image_Lab_Activity in Visual Studio Code.
2. Inside it, create index.html and style.css. Add an images folder for four image files.
3. Paste the starter HTML and CSS from this sheet. Use teacher-provided images or images you have permission to use; update each src and href path.
4. Open index.html in a browser. If an image does not appear, check the filename, extension, and folder path.

PART B — Build and style the gallery

5. Confirm each image is inside a .gallery-card container and has meaningful alt text.
6. Set .gallery to display: flex and flex-wrap: wrap. Add a gap between cards.
7. Style each card with a border and consistent width. Set each image to width: 100% and height: auto.
8. Add a caption under each image and center its text.
9. Create a :hover effect that changes the card border color.
10. Add media queries: four cards across on wide screens, two across on medium screens, and one across on narrow screens.

PART C — Understand image sprites

11. An image sprite combines several small graphics into one larger image file.
12. Use background-image to load the sprite and background-position to reveal the needed section.
13. Set an element's width and height to the dimensions of the icon so other portions remain hidden.
14. Inspect the sample positions: Home uses 0 0, Previous uses -47px 0, and Next uses -91px 0.
15. Replace the sample sprite URL with the correct local or teacher-provided file. Icons will not display if the sprite file is missing.

PART D — Test and improve

16. Resize the browser window to test all responsive breakpoints.
17. Check that captions are readable and images keep their proportions.
18. Use browser developer tools to inspect CSS rules that are not working.
19. Add a short heading and one-sentence introduction to make the page your own.
20. Save your work and take a screenshot showing the gallery and sprite navigation.

Mini-Project — TechPro Photo Showcase

Create a polished mini-gallery featuring four school-appropriate topics, such as campus spaces, local scenery, school activities, or nature. Include:

- A page title and short introduction.
- Four image cards with meaningful alt text and captions.
- A hover effect and responsive layout.
- A sprite-based navigation strip with Home, Previous, and Next icons using a sprite file supplied or approved by your teacher.
- A brief note (2–3 sentences) explaining how the gallery responds to screen sizes and how sprite positions work.

Challenge Task

Improve the design with consistent spacing, a subtle card shadow, and a clear keyboard-focus style. Keep the page readable on a small screen and retain descriptive alt text.

Submission Checklist

☐ index.html and style.css are saved in the project folder.

☐ Four gallery items include images and descriptive alt text.

☐ Each image has a caption and scales to its card width.

☐ Gallery cards have a visible border and hover effect.

☐ Responsive layout works at wide, medium, and narrow widths.

☐ Sprite icons use background-image, width/height, and background-position.

☐ I tested the page and captured a screenshot.

Review Questions

1. What is the purpose of a CSS image gallery?
   Answer: organizes multiple images and captions into a clean, structured layout grid on a webpage

2. What is the role of the .gallery-card container?
   Answer: groups each individual image and its corresponding caption together as a single styled card unit

3. Why is width: 100% useful for an image inside a card?
   Answer: ensures the image automatically scales to completely fill the width of its container card while remaining responsive

4. What does the :hover pseudo-class do in this activity?
   Answer: changes the gallery card's border color when the user places their mouse pointer over it

5. What is a responsive image gallery?
   Answer: gallery that automatically adjusts its layout columns and card widths based on different screen sizes and device

6. What does a media query allow CSS to do?
   Answer: allows CSS to apply different styling rules depending on specific device characteristics, such as maximum or minimum screen widths

7. What is an image sprite?
   Answer: a single image file that combines multiple smaller images, allowing web pages to load faster

8. Which CSS property loads a sprite image?
   Answer: Background-image property

9. How does background-position help show one icon from a sprite?
   Answer: it shifts the coordinates of the background image to display only the specific portion or icon needed while hiding the rest

10. Name one troubleshooting step if a gallery image does not appear.
    Answer: check if the file path is correct

Quick Review

Gallery layout: Containers group each image and caption; flex layout and wrapping arrange cards.

Image sizing: width: 100% fills the card; height: auto helps preserve proportions.

Responsive layout: Media queries apply CSS rules at specified screen widths.

Image sprite: One image file contains multiple small graphics.

Sprite positioning: background-image loads the sprite; background-position selects the portion to display.

References

W3Schools — CSS Image Gallery: https://www.w3schools.com/css/css_image_gallery.asp

W3Schools — CSS Image Sprites: https://www.w3schools.com/css/css_image_sprites.asp

Reminder: img_navsprites.gif must be available in your project folder or replaced with an approved image path. Image files are not embedded in this activity sheet.